Amriteshkumar Yadav

Data Scientist

Ahmedabad, India | amriteshyadav2004@gmail.com | +91 8320154269 | github.com/Amrit-04 |
linkedin.com/in/amriteshkumar-yadav-214594250

Education

SRM Institute of Science and Technology — Chennai, India
B.Tech in Computer Science and Engineering (AI/ML Specialization), CGPA: 7.98/10 — *Graduated Jul 2026*

Technical Skills

Languages: Python, SQL, R, C++, C, Go, JavaScript, TypeScript, Bash
Data Science & ML: PyTorch, Scikit-learn, XGBoost, LightGBM, HMM, GNNs, Walk-Forward CV, Monte Carlo
LLMs & Vision: Llama 3 (Ollama), RAG, LangChain, Cross-Encoders, Ragas, ChromaDB, OpenCV, YOLO
Data Engineering & MLOps: Apache Kafka, PostgreSQL, Redis, MLflow, ONNX Runtime, CUDA, Docker, Linux, AWS

Experience

Software Engineering Intern — Jul 2025 – Jan 2026
PopFix (now SuckSi) – University-Incubated Nutraceutical Startup — *Remote*

- Spearheaded **full-stack web development** (**React.js, Next.js**) and cloud setup as the **first technical hire**, configuring production domains, DNS routing, and automated order-processing workflows from scratch.
- Developed and deployed **Python automation** scripts for order tracking, vendor coordination, and logistics workflows, reducing manual processing time by **30%** and scaling operations to **100+ monthly orders**.
- Partnered with founders in go-to-market strategy meetings as the **sole technical voice**, translating business requirements into standardized fulfillment workflows that eliminated manual spreadsheet re-keying.

Projects

Moneyball: Quantitative Portfolio Analytics & GPU Risk Simulation Engine GitHub
Python, PyTorch (CUDA), Scikit-learn, XGBoost, LightGBM, Hidden Markov Models (HMM), NumPy, Parquet

- Built a cross-sectional alpha pipeline across 24 assets (1,260 trading days) using **purged walk-forward cross-validation** (504d train, 5d purge/embargo), reaching **0.0359 out-of-sample Rank IC** and **1.521 annualized ICIR**.
- Developed a **3-state Gaussian HMM** on log-returns and volatility to scale regime exposure, cutting 3-year out-of-sample **max drawdown from 13.58% (1/N baseline) to 8.59% (2.02 Sharpe)**, net of 10 bps turnover costs.
- Architected a **PyTorch CUDA Monte Carlo engine** simulating **1M Jump-Diffusion paths in 29.26 ms (42.9x speedup over NumPy)** for 99% VaR/CVaR, plus a **14-qubit QAOA/QUBO optimizer** reaching a **95.49% approximation ratio**.

Neural Database Query Optimizer (Graph Neural Networks) GitHub
Python, PyTorch, GNNs (GCN, GAT, GraphSAGE), ONNX Runtime, PostgreSQL 16 (TPC-H), Scikit-learn, MLflow

- Trained **Graph Neural Network (GCN, GAT, GraphSAGE)** cost models in PyTorch using pairwise margin ranking loss on 184 PostgreSQL 16 `EXPLAIN` plans (12 TPC-H templates), reaching **0.74 Spearman correlation**.
- Evaluated zero-shot plan selection via **leave-one-template-out cross-validation**, outperforming gradient boosting by **+9.7 percentage points (38.9% vs. 29.2% top-1)** and achieving a **2.17x speedup** on TPC-H Q10.
- Tracked **12 architecture and pooling ablations in MLflow** and exported the winning GCN model to **ONNX Runtime**, delivering **0.21 ms p95** single-thread CPU inference (**0.44 ms p95 end-to-end** with graph parsing).

LLM Reliability Control Plane & RAG Evaluation Gateway GitHub
Python, C++, Llama 3.2 (Ollama), LangChain, ChromaDB, Redis, FastAPI, OpenTelemetry (OTLP), DuckDB, Docker

- Built a **local RAG and hallucination guardrail pipeline** (**Llama 3.2 4-bit Q4_K_M**, ChromaDB, Pydantic) using Cross-Encoder verification, semantic similarity, and circuit-breaker fallbacks to block ungrounded outputs.
- Engineered an **OpenTelemetry (OTLP) trace ingestion gateway** in C++ and Python with CRC-32 validation and 64 MB WAL rotation, achieving **5 ms p99 ingest latency** (<**50 MB memory**) across **54 automated tests**.
- Implemented a **2-phase Zstd Parquet compaction** pipeline sustaining **50,000 spans/sec**, paired with a **Redis semantic cache** (90% cosine threshold) and **DuckDB analytical queries** completing in <**100 ms p99**.

Publications

IEEE RMKMATE 2026: *"Workplace Safety Using Computer Vision: A Framework for Personal Protective Equipment Detection and Intrusion Monitoring"* (IEEE Xplore: 11518869 | GitHub) – Co-authored a real-time YOLO and OpenCV pipeline for PPE compliance and restricted-zone intrusion tracking.

Certifications & Achievements

Oracle Generative AI Professional (2025) | Red Hat Certified System Administrator (RHCSA) | Oracle OCI Foundations Associate | Quantum Quest Expo 2024 Finalist